💼 Use Case:

Let users ask questions in plain English like “Show me sales from Mumbai last month” and get real answers from the database—no SQL needed.

💭 Mindset / Thought Process

1. I’ll create something that the user can chat with using natural language.
2. I need something that converts my natural language to SQL query.
3. I need something that can run the SQL query directly in Database and show results in chat.
4. I’ll test it multiple times by giving complex questions.

🧩 Solution Design

- Something that starts the flow when I send a message.
- Some AI that understands what I meant and plans how to answer.
- A way for the AI to look at all the tables in the database, what's inside each table (like columns), and the ability to run the SQL query and fetch the data

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🪜 Step-by-Step Execution

🪜 Step 1: Set up your N8N Account

Get a trial account (14 days) here - https://app.n8n.cloud/register

(or)

Set up a local hosting setup, follow the guide attached in the lecture.

🪜 Step 2: Get the automation steps using Perplexity / ChatGPT and use them for asking questions at every step

Link - https://chatgpt.com/share/6864baf6-1b0c-8012-b95d-dd59afcb61c0

Prompt:

I want to create an AI Agent in n8n that helps in writing SQL queries, executing them in my Postgres DB, and returning the answers whenever we ask questions in plain English. It should also check if there is actual data

available to answer that question; otherwise, it should ask follow-up questions and then answer. If there is no data available to answer the question, it should clearly mention, 'There is no data available to answer your questions. Here are the different details I can help you with. Draft the steps for this automation.

🪜 Step 3: Connect with Trigger (chat)

- Watches the Incoming message and triggers the AI Agent.

🪜 Step 4: Add an AI Agent Node

- This AI Agent will be our smart Junior Analyst with SQL knowledge
  - Use the system message we got from the first prompt response, if not create it using the following prompt.
  - Prompt:
    Draft a system message for the AI agent, which needs to act as a SQL Agent, and turn the user's plain English queries into SQL queries. It should be smart enough to understand what tables are available in our Postgres DB. If the question is not clear, it should ask follow-up questions for confirmation and then proceed with answering the questions. In case of unavailability of data to answer the particular question, it should specifically mention, 'There is no available data to answer the particular question, but I can help with the following details.

🪜 Step 5: Give Power and Life to the AI Agent

- Attach the necessary tools the agent needs to work with the Postgres DB:
  - To check the table structure and schema
    - Prompt (use it in ChatGPT): *Help me with the SQL query that can retrieve the schema details from the Postgres DB*
  - To run the final query
- Add a brain (LLM like Gemini) and memory to help it think and remember the conversation.

🪜 Step 6: Test the Agent

- Ask real business questions.
- Make sure it generates the correct query and gives the right answer.
- Perform UAT to validate edge cases and error handling.

✅ **Final Outcome**

- The user chats with an AI and gets answers from the database.
- The AI handles the thinking, querying, and explaining.
- Teams get insights instantly, without depending on the tech person.